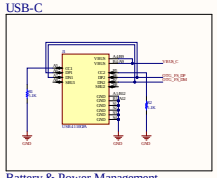
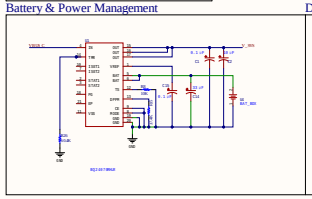
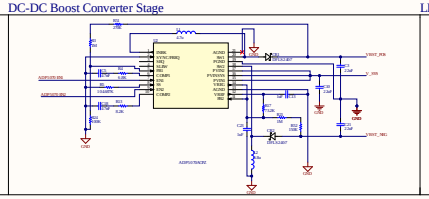
USB-C



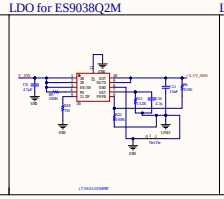
Battery & Power Management



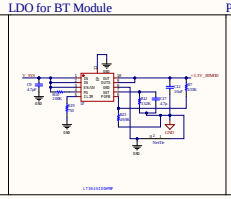
DC-DC Boost Converter Stage



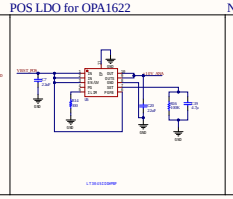
LDO for ES9038Q2M



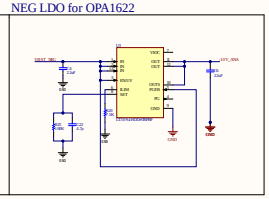
LDO for BT Module



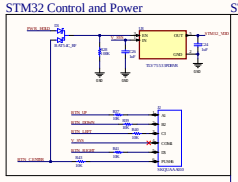
POS LDO for OPA1622



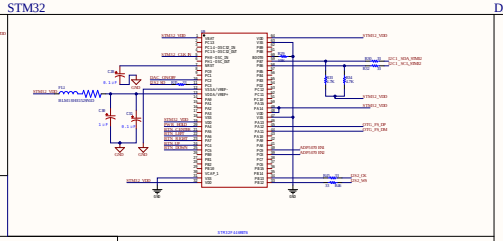
NEG LDO for OPA1622



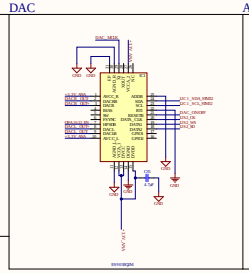
STM32 Control and Power



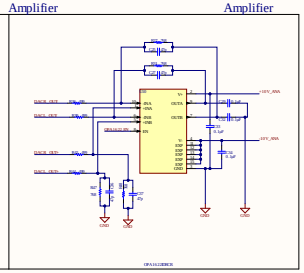
STM32



DAC

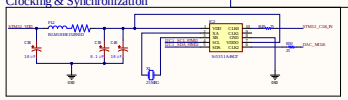


Amplifier



Amplifier

Clocking & Synchronization



Schematic	Schematic1	Create at	2025-10-09
		Update at	2026-01-30
Board	Board1	Page	P1
Drawn	onear_headphones		
Reviewed			
		Version	Size	Page 1 Total 1
		V1.0	A4	EasyEDA.com

Line #	Name	Description	Designator	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	0.1µF		C1	1							
	10µF		C2	1							
	Capacitor	Capacitor	C3, C4, C6, C7, C10, C20, C21	7	Murata	GRM32ER71H106KA12L	Volume Production	Arrow Electronics	GRM32ER71H106KA12L	0,3643	364,3
	Capacitor	Capacitor	C5, C18	2	Murata	GRM155R71H473KE14D	Volume Production	Farnell	3582817	0,0119	0,119
	Capacitor	Capacitor	C8, C9	2							
	Capacitor	Capacitor	C11, C12	2							
	Capacitor	Capacitor	C13, C23, C24, C26	4							
	33µF		C14	1	KEMET	T495D336K016ATE225	Volume Production	Arrow Electronics	T495D336K016ATE225	0,1318	0,1318
	0.1µF		C15, C28, C31, C39	4	Murata	GRM155R71H104KE14D	Volume Production	Farnell	2611912	0,0146	0,146
	Capacitor	Capacitor	C16, C17, C19, C22	4							
	Capacitor	Capacitor	C25, C27, C36, C37	4	Murata	GCM1555C1H470JA16D	Volume Production	Future Electronics	5120075	0,0028	28
	Capacitor	Capacitor	C29, C32, C33, C34	4	Murata	GRM155R71H104KE14D	Volume Production	Farnell	2611912	0,0146	0,146
	1µF		C30	1							
	Capacitor	Capacitor	C35	1	Samsung Electro-Mechanics	CL10B475KQ8NQNC	Volume Production	Avnet	82AC3361	0,134	0,134
	10µF		C38, C40	2							
	DFLS240-7	Description	CR1, CR2	2	Diodes Inc.	DFLS240-7	Volume Production	Mouser	621-DFLS240-7	0,77	1,54
	BAT54C_RF	SOT-23, 30V, 0.2A, Schottky Diode	D1	1	Taiwan Semiconductor	BAT54CRF	Volume Production				
	BLM18HE152SN1D	1.5 kOhms @ 100MHz 1 Signal Line Ferrite Bead 0603 (1608 Metric) 500mA 500mOhm	FL1, FL2	2	Murata	BLM18HE152SN1D	Volume Production	Mouser	81-BLM18HE152SN1D	0,18	0,36
	ES9038Q2M	Integrated Circuit	IC1	1	ESS	ES9038Q2M	Unknown	Mouser	46D-ES9038Q2M	19,2	19,2
	SI5351A-B-GT	Integrated Circuit	IC2	1	Skyworks Solutions	SI5351A-B-GT	Volume Production	Mouser	634-SI5351A-B-GT	2,11	2,11
	USB4110GFA	USB-C (USB TYPE-C) USB 2.0 Receptacle Connector 24 (16+6 Dummy) Position Surface Mount, Right Angle	J1	1	Global Connector Technology	USB4110-GF-A	Volume Production	Farnell	3588206	1,18	1,18
	SKQUAAA010		J2	1	MARSHALL_DONOR	1	Manual Solution	ME	1	1	1
	Inductor	Inductor	L1	1	Würth Elektronik	74438357047	Unknown	Farnell	2842186	1,45	1,45
	Inductor	Inductor	L2	1	Würth Elektronik	74438357068	Unknown	Farnell	2842188	1,43	1,43
	Resistor	Resistor	R1, R2	2	Yageo Group	RC0402FR-075K1L	Volume Production	Mouser	603-RC0402FR-075K1L	0,005	0,05
	Resistor	Resistor	R3, R21	2	RALEC	RTT061004BTP	Manual Solution	LCSC	C312263	0,0514	0,1028
	Resistor	Resistor	R4	1	Yageo Group	RT0402BRD076K8L	Volume Production	Mouser	603-RT0402BRD076K8L	0,29	0,29
	Resistor	Resistor	R5	1	Yageo Group	RT0402BRD07105KL	Volume Production	Mouser	603-RT0402BRD07105KL	0,15	0,15
	Resistor	Resistor	R6, R7	2	CHANGLONG	CL0402FN470KP	Manual Solution	LSCS	C46635183	0,0007	0,0014
	Resistor	Resistor	R8, R29, R37, R39, R40, R41, R43	7	Yageo Group	RC0402FR-0710KL	Volume Production	Mouser	603-RC0402FR-0710KL	0,007	0,07
	Resistor	Resistor	R9, R10	2	Yageo Group	RC0402FR-07200KL	Volume Production	Farnell	1458761	0,0031	0,031
	Resistor	Resistor	R11, R12	2	Panasonic	ERA-2AEB3322X	Volume Production	Mouser	667-ERA-2AEB3322X	0,1	0,2
	Resistor	Resistor	R13	1	Yageo Group	RT0402BRD078K2L	Volume Production	Mouser	603-RT0402BRD078K2L	0,1	0,1
	Resistor	Resistor	R14	1	Panasonic	ERA-2AEB301X	Volume Production	Mouser	667-ERA-2AEB301X	0,1	0,1
	Resistor	Resistor	R15	1	Panasonic	ERA-2AEB3742X	Volume Production	Arrow Electronics	ERA-2AEB3742X	0,0084	0,0084
	Resistor	Resistor	R16, R24, R25, R28	4	Panasonic	ERA-2AEB104X	Volume Production	Mouser	667-ERA-2AEB104X	0,1	0,4
	Resistor	Resistor	R17	1	Panasonic	ERA-2AEB7322X	Volume Production	Mouser	667-ERA-2AEB7322X	0,1	0,1
	Resistor	Resistor	R18, R19	2	Panasonic	ERA-2AEB751X	Volume Production	Arrow Electronics	ERA-2AEB751X	0,019	0,038
	Resistor	Resistor	R20	1	Yageo Group	RT0402BRD077K5L	Volume Production	Arrow Electronics	RT0402BRD077K5L	0,0252	0,0252
	Resistor	Resistor	R22, R23	2	Panasonic	ERA-2AEB4992X	Volume Production	Farnell	2563607	0,062	0,62
	Res_0603	Resistor	R26	1	Yageo Group	RC0402FR-0760K4L	Volume Production	Farnell	3495651	0,0069	0,069
	Resistor	Resistor	R27, R31, R47, R48	4	Panasonic	ERA2AEB7680X	Volume Production	Mouser	667-ERA-2AEB7680X	0,1	0,4
	Resistor	Resistor	R30, R32, R35, R45, R46	5	Yageo Group	RT0402BRD0733RL	Volume Production	Mouser	603-RT0402BRD0733RL	0,047	0,47
	Resistor	Resistor	R33, R34	2	Yageo Group	RC0402FR-074K7L	Volume Production	Mouser	603-RC0402FR-074K7L	0,004	0,04
	Resistor	Resistor	R36, R38, R42, R44	4	Panasonic	ERA-2AEB4990X	Volume Production	Mouser	667-ERA-2AEB4990X	0,1	0,4
	Resistor	Resistor	R49, R50	2	Yageo Group	RT0402BRD0724R8L	Volume Production	Mouser	603-RT0402BRD0724R8L	0,2	0,4
	Resistor	Resistor	R51	1	KOA	RS73F2BTTD2703B	Manual Solution	LCSC	C4216782	0,0978	0,0978
	Resistor	Resistor	R52	1	FOJAN	FRHD805B15037S	Manual Solution	LCSC	C5204040	0,014	0,014
	BQ24070RHLR		U1	1	Texas Instruments	BQ24070RHLR	Volume Production	Mouser	595-BQ24070RHLR	2,72	2,72
	ADP5070ACPZ	1 A/0.6 A, DC to-DC Switching Regulator with Independent Positive and Negative Outputs	U2	1	Analog Devices	ADP5070ACPZ	Volume Production	Newark	ADP5070ACPZ	6,1	6,1
	LT3094HDD#TRPBF	â¯ 20V, 500mA, Ultralow Noise, Ultrahigh PSRR Negative Linear Regulator	U3	1	Analog Devices	LT3094HDD#TRPBF	Volume Production	Mouser	584-LT3094HDD#TRPBF	13,63	13,63
	BAT_BOX		U4	1	ME	1	Manual Solution	ME	1	1	1
	LT3045IDD#PBF		U5, U6, U7	3	Analog Devices	LT3045IDD#TRPBF	Volume Production	Newark	LT3045IDD#TRPBF	10,76	32,28
	TLV75533PDBVR	500-mA High-PSRR Low-IQ-Tief-Dropout-Spannungsregler mitablem	U8	1	Texas Instruments	TLV75533PDBVR	Volume Production	Mouser	595-TLV75533PDBVR	0,36	0,36
	STM32F446RET6		U9	1	STMicroelectronics	STM32F446RET6	Volume Production	Arrow Electronics	STM32F446RET6	4,35	4,35
	OPA1622IDRCR	SoundPlusâ¿¢ Audio Operational Amplifier with High Performance, Low THD+N and Bipolar Input	U10	1	Texas Instruments	OPA1622IDRCR	Volume Production	Mouser	595-OPA1622IDRCR	6,62	6,62
	Crystal	Crystal	X1	1	NDK	NX3225SA-25.000M-STD-CRS-2	Volume Production	Mouser	344-NX3225SA25MSTRS2	1,08	1,08